MIL-D-51161A(EA)
17 May 1978
SUPERSEDING
MIL-D-51161(MU)
18 November 1963

MILITARY SPECIFICATION

N,N'-DIISOPROPYLCARBODIIMIDE

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one grade of N,N'-diisopropylcarbodiimide.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-D-729 - Drums, Shipping and Storage, 55-Gallon (208 Liters).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS

49 CFR 171 to 179 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Assay. N,N'-Diisopropylcarbodiimide shall contain no less than 95.0 percent by weight N,N'-diisopropylcarbodiimide when tested as specified in 4.2.4.1.

3.2 Sulfur. N,N'-Diisopropylcarbodiimide shall contain no more than 0.1 percent by weight sulfur when tested as specified in 4.2.4.2.

3.3 Infrared absorption characteristics. When tested as specified in 4.2.4.3, the infrared spectrum of the N,N'-diisopropylcarbodiimide shall show the following bands:

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(a) An intense sharp absorption band with maximum between 4.67 and 4.77 micrometers,

(b) A weak, double absorption band with maxima near 7.22 and 7.33 micrometers, and

(c) A medium intensity absorption band near 7.63 micrometers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the N,N'-diisopropylcarbodiimide offered for acceptance at one time which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For exmination of preparation for delivery. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test (see 6.4). Sampling for test shall be conducted in accordance with table I. A representative specimen of approximately 0.5 liter shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE I. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled and closed drum ready for shipment. Sample drums shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Plug closures not as specified or incorrectly torqued
(d) Closure cap-seal not as specified or missing
(e) Container damaged or leaking
(f) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests (see 6.4). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Assay.

(a) 1,4-Dioxane. Reflux 1,4-dioxane over calcium hydride and distill to 0.05 percent or less water. Store in a desiccator.

(b) Oxalic acid solution. Dry oxalic acid in a vacuum at 50° ± 2° C for 4 hours, breaking the crust every hour. Prepare a 0.1N solution of the oxalic acid in the 1,4-dioxane prepared in (a).

(c) Procedure. Weigh 0.070 to 0.150 g of the specimen in a micro weighing bottle or glass ampoule. Place the bottle or ampoule in a dry 250-milliliter (ml) Erlenmeyer flask containing 40 ml of the oxalic acid solution prepared in (b). If the container is a glass ampoule, crush it with a glass rod to release contents and wash down the rod with 5 ml of the 1,4-dioxane prepared in (a). If a weighing bottle is used, remove the cover as soon as the bottle has been placed in the oxalic acid solution and wash down the implement used in removing the cover with 5 ml of the 1,4-dioxane prepared in (a). Reflux the contents of the flask for 15 minutes on a hot plate. Wash down the condenser with 10 to 15 ml of

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water. Remove the flask from the hot plate, add 100 ml of water, and titrate with standardized 0.1N sodium hydroxide solution to the first pink color using 1 ml of a 1-percent solution of phenolphthalein in ethanol as the indicator. Run a blank on 40 ml of the oxalic acid solution prepared in (b) using the same conditions as in the specimen test. Calculate the percent by weight N,N'-diisopropylcarbodiimide as follows:

$$\text{Percent N,N'-diisopropylcarbodiimide} = \frac{6.31(A - B)C}{W}$$

where: A = Milliliters of sodium hydroxide solution required to titrate the blank,
B = Milliliters of sodium hydroxide solution required to titrate the specimen,
C = Normality of the sodium hydroxide solution, and
W = Weight of specimen in grams.

4.2.4.2 Sulfur. Place 5 g of the specimen, weighed to the nearest 0.1 g, in a 50-ml nickel crucible containing 10 g of flux (1 part sodium peroxide to 3 parts sodium carbonate). Add an additional 10 g of flux, stir until thoroughly mixed, and cover with a thin layer of sodium carbonate. Heat the crucible carefully on a hot plate in a fume hood until the evolution of copious white fumes subsides. Avoid breathing these fumes. (Warning: Application of too much heat initially may result in an uncontrolled oxidation reaction. Explosive expulsion of the hot contents of the crucible may occur if this warning is not heeded.) Heat the crucible over a bunsen flame until the flux becomes liquid, then place the crucible in a muffle furnace preheated to 400° C. Gradually raise the temperature to 800° C and maintain this temperature until fusion is complete (about 1/2 hour). Remove the crucible from the furnace and cool to room temperature. Dissolve the melt from the crucible in a 600-ml beaker containing about 300 ml of nearly boiling water. After the melt has dissolved, remove the crucible from the solution and wash it down with water. After allowing the solution to cool to near room temperature, neutralize with concentrated hydrochloric acid and add 1 ml excess for each 200 ml of solution. Filter through an unwashed, dense filter paper. If the filtrate appears cloudy, refilter. Heat the filtrate to boiling and add 10 ml of 10-percent barium chloride solution. Heat on a steam bath for at least 1 hour. Remove from the steam bath and allow to stand for 3 hours. Filter through a prepared and ignited Gooch crucible which has been weighed to the nearest 0.1 milligram. Wash the precipitate thoroughly with water until it is free of chloride. Ignite the precipitate in a muffle furnace to constant weight, cool to room temperature, and weigh. Calculate the percent by weight sulfur as follows:

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$$\text{Percent sulfur} = \frac{13.73A}{W}$$

where: A = Weight of residue in grams and
W = Weight of specimen in grams.

4.2.4.3 Infrared absorption characteristics. Prepare a 1-percent by volume solution of the specimen in carbon tetrachloride. Fill a sealed infrared absorption cell of approximately 0.1 millimeter thickness with the solution and place it in a double beam infrared spectrophotometer. Place a matched cell containing carbon tetrachloride in the reference beam. Record the spectrum from 2.5 to 8 micrometers.

5. PACKAGING

5.1 Packing, level A. A specified quantity (see 6.2) of N,N'-diisopropyl carbodiimide + 1 or - 0 percent by weight shall be packed in a nominal 208.2-liter capacity steel drum conforming to type II of PPP-D-729. The quantity shall not completely fill the drum at a temperature of 54.4° C. Gasket and seaming compounds shall neither affect nor be affected by the N,N'-diisopropylcarbodiimide. The 19.05-millimeter plug and the 50.8-millimeter plug shall be torqued closed to 20 and 40 Newton-meters, respectively. Each closure shall be furnished with a tamper-resistant cap seal.

5.2 Marking. Shipments shall be marked in accordance with Department of Transportation regulations and MIL-STD-129.

6. NOTES

6.1 Intended use. N,N'-Diisopropylcarbodiimide is intended for use as a chemical stabilizer.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Weight of chemical per drum required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing precaution. N,N'-diisopropylcarbodiimide is highly toxic and causes corneal edema and dermatitis. Rat inhalation studies show that LC_{50} is 20 to 25 parts per million for a 4-hour exposure.

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Because this specification covers sampling and testing of material which is potentially hazardous to personnel, it is emphasized that all applicable safety rules, regulations, and procedures must be followed in handling and processing this material.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practice for Indicating Which Places of Figures Are To Be Considered Significant in Specified Limiting Values (ASTM E29).

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Project No. 6810-A817

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-D-5716A 17 MAY 77

N,N'-DIISOPROPYLCARBODIIMIDE

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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